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On the (almost) global exponential convergence of overparameterized policy optimization for the LQR Problem

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***Abstract*— In this work we study the convergence of gradient methods for nonconvex optimization problems – specifically the effect of the problem formulation on the convergence behavior of gradient flow solutions. We show through a simple example that, surprisingly, the gradient flow solution can be exponentially or asymptotically convergent, depending on how the problem is formulated. We then deepen the analysis and show that a policy optimization strategy for the continuous-time linear quadratic regulator (LQR) (which is known to present only asymptotic convergence globally) presents almost global exponential convergence if the problem is overparameterized through a linear feed-forward neural network (LFFNN). We prove that this qualitative improvement always happens for a simplified version of the LQR problem and derive explicit convergence rates for the gradient flow. Finally, we show that both the qualitative improvement and the quantitative rate gains persist in the general LQR problem through numerical simulations.**

I. Introduction

Motivated by the rise in popularity of machine learning and neural networks, research on gradient methods, policy optimization and their application have been the focus of much attention in recent years [1]–[6]. The intuitive idea of finding the minimum of a function by "traveling along" its direction of steepest descent stood the test of time, with many recent studies [7]–[11] exploring inherent properties that justify the strength of gradient methods.

Despite this success, gradient methods are not guaranteed to solve (or even converge for) every problem, and additional assumptions are usually required. In [12] the authors provide an overview of different assumptions in optimization theory, as well as their relationship. Of particular note is the assumption of convexity, which results in its own field of research in optimization theory [13].

Alternatively, the Polyak-Łojasiewicz inequality (PŁI) [12], [14] provides a more general guarantee of convergence for solutions of gradient methods, being one of the most important assumptions in nonconvex optimization. Despite that, however, some problems can be shown to not satisfy any PŁIglobally, but still be convergent and optimal under gradient methods.

An interesting concrete example of this can be found when studying policy optimization formulations for the linear quadratic regulator (LQR) problem [5], [9], [15]–[21]. In [5] the authors showed that a policy optimization method for the discrete-time LQR problem can be shown to always satisfy a PŁIglobally, however existing results for the continuous-time version of the problem [9], [16] were only able to show that a PŁIholds on any level set of the cost function.

At first glance, the distinction might appear irrelevant, since properness of the LQR cost function implies that any initialization is contained in some of its level set. However, this distinction motivated follow-up research [9], [22]. In [9] the authors first showed its importance by proving distinct robustness properties for problems satisfying either condition. In the same paper the authors also characterized a stronger "type" of PŁIinequality for the continuous-time LQR policy optimization problem. In [22] we formally introduced different types of PŁIand illustrated their relationship and different convergence guarantees. Of particular interest to this paper is the relationship between the usual PŁI(named global PŁIor *g*PŁIin [22]) and the saturated PŁI(*sat*PŁI). In [22] we show that any cost satisfying a *g*PŁIsatisfies a *sat*PŁI, however the converse is not true. We further prove that gradient flows of problems satisfying only a *sat*PŁIpresent qualitatively worse convergence properties than those satisfying a *g*PŁI. This illustrates the importance in the distinction between these concepts, and why one might be interested in their relationship.

In this paper, we focus on the *sat*PŁIand *g*PŁIand their dependence on the problem formulation. Formulation-based acceleration of gradient methods is a well-known phenomenon in the literature, with linear feedforward neural networks (LFFNN) having results in the literature on initialization-dependent quantitative improvements on the convergence rate of gradient methods [23]–[25]. However the question we investigate in this paper regards a possible qualitative improvement on the convergence of solutions.

We begin by formally introducing the concepts in section II, and then illustrate that a simple reparameterization of the form $f(k) := \mathcal{L}(k^2)$ can convert a function $\mathcal{L}(k)$ that satisfies only a *sat*PŁI(and thus presents only asymptotic convergence to gradient methods) into a function $f(k)$ that can provably be shown to satisfy a *g*PŁIalmost everywhere. In section III, we return to the policy optimization continuous-time LQR as a testbed, and explore the effects of using a LFFNN to the convergence of solutions of a gradient flow. We prove that, for a simplified case of the problem, the overparameterized problem formulation (i.e. using the LFFNNs) improves the convergence of solutions from linear–exponential to exponential, with the trade-off of adding a saddle-point to the dynamics, which can trap

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or slow-down some solutions. We then provide, in section IV, numerical evidence of this same observation holding for general LQR problem, illustrating the properties discussed theoretically in the previous section. We finish the paper in section V with an overview of our results and some possible future directions of work related to this paper. Finally, we provide proof sketches in this paper and defer the full proofs to the extended arXiv version [26].

II. Theoretical Framework

A. Preliminaries

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be open and $\mathcal{L} : \mathcal{X} \to \mathbb{R}$ be a real-analytic, bounded below, and proper (i.e. coercive or radially unbounded) loss/cost function. We consider the optimization problem

$$\min_{k\in\mathcal{X}} \ \mathcal{L}(k). \tag{1}$$

which we assume is nonempty. Let $\underline{\mathcal{L}} := \inf_{k\in\mathcal{X}} \mathcal{L}(k)$ and let $\mathcal{T} := \{k \in \mathcal{X} \mid \mathcal{L}(k) = \underline{\mathcal{L}}\}$ denote the *target set*, i.e., the set of points that solve (1). Define the optimality gap $\theta(k) := \mathcal{L}(k) - \underline{\mathcal{L}} \geq 0$ and consider the gradient flow

$$\dot{k}(t) = -\nabla\mathcal{L}(k(t)), \qquad k(0) \in \mathcal{X}, \tag{2}$$

for finding a solution of (1). One can verify that along the gradient flow, $\dot{\theta}(t) = -\|\nabla\mathcal{L}(k(t))\|^2$. Since $\mathcal{L}$ is smooth on the open set $\mathcal{X}$, every global minimizer is stationary; hence $\mathcal{T} \subseteq \mathcal{Z} := \{k \in \mathcal{X} : \nabla\mathcal{L}(k) = 0\}$.

Despite this, note that stationarity is, in general, only a necessary condition for optimality, and convergence of a gradient method to global minima requires additional structure. Ideally, one would characterize the following property of the gradient flow (2):

Definition 1 (global exponential cost stability (GECS))**.** *Let $\phi : \mathbb{R}_+ \times \mathcal{X} \to \mathcal{X}$ denote the solution of* (2) *for any initial condition in $\mathcal{X}$. The gradient flow* (2) *is said to be* GECS *if there exists $\mu > 0$ such that*

$$\theta\big(\phi(t, k_0)\big) \leq \theta(k_0)\, e^{-\mu t}, \qquad \forall t \geq 0,\ \forall k_0 \in \mathcal{X}. \tag{3}$$

The following definition and lemma introduce conditions that are necessary and sufficient for this property to hold.

Definition 2 (global Polyak–Łojasiewicz (gPŁI))**.** *A function $\mathcal{L}$ satisfies a global Polyak–Łojasiewicz* (gPŁI) *if there exists a $\mu > 0$ for which the inequality holds:*

$$\|\nabla\mathcal{L}(k)\| \geq \sqrt{\mu(\mathcal{L}(k) - \underline{\mathcal{L}})}, \qquad \forall k \in \mathcal{X}. \tag{4}$$

Lemma 1 (Lemma 2 of [22])**.** *The gradient flow* (2) *of the minimization problem* (1) *is* GECS *if and only if its loss function satisfies a* gPŁI.

Traditionally, the gPŁI (or just PŁI as it is usually referred to in the literature [12], [14]) is a powerful tool in nonconvex optimization, linking the gradient norm $\|\nabla\mathcal{L}(k)\|$ to the gap $\theta(k)$ and thus quantifying exponential decay rates for solutions of (2). The gPŁI, however, can be impossible to characterize for some problems, and weaker versions of the inequality may be sufficient to provide weaker convergence guarantees. An example is the saturated PŁI condition, defined as follows:

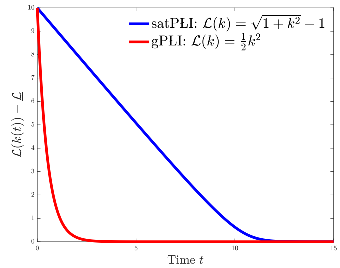


Fig. 1: Depiction of two gradient flow solutions, one whose cost satisfies a gPŁI, and one whose cost satisfies only a *sat*PŁIand with a globally bounded gradient.

Definition 3 (Saturated PŁI (*sat*PŁI))**.** *A function $\mathcal{L}$ satisfies a saturated* PŁI (*sat*PŁI) *if there exist $a, b > 0$ for which*

$$\|\nabla\mathcal{L}(k)\| \ \geq \ \sqrt{\frac{a(\mathcal{L}(k) - \underline{\mathcal{L}})}{b + (\mathcal{L}(k) - \underline{\mathcal{L}})}}, \qquad \forall k \in \mathcal{X}. \tag{5}$$

In [9] the authors showed that a policy-optimization strategy for the continuous-time linear quadratic regulator (LQR) satisfies a *sat*PŁI as defined above. Furthermore, in [22], we showed that the same problem will never satisfy a gPŁI.

A priori, if the cost satisfies a *sat*PŁI, then only asymptotic convergence can be guaranteed. However, two asymptotically convergent solutions can differ greatly in their qualitative behavior, and a specific type of convergence behavior can be characterized for cost functions that satisfy a *sat*PŁI.

Definition 4 (global linear–exponential cost stability (GLECS))**.** *Let $\phi : \mathbb{R}_+ \times \mathcal{X} \to \mathcal{X}$ be the solution of* (2) *for any initial condition in $\mathcal{X}$. The gradient flow* (2) *is said to be* GLECS *if there exists $\beta > 0$ such that, for every $k_0 \in \mathcal{X}$ and every $t^* \geq 0$, there exists $\mu_{k_0,t^*} > 0$ for which*

$$\theta(\phi(t, k_0)) \leq \begin{cases} \theta(\phi(t^*, k_0)) - \beta(t - t^*), & t \leq t^*, \\ \theta(\phi(t^*, k_0))\, e^{-\mu_{k_0,t^*}(t-t^*)}, & t > t^*. \end{cases} \tag{6}$$

Lemma 2 (Lemma 3 of [22])**.** *If the gradient flow* (2) *of the minimization problem* (1) *satisfies a* satPŁI (5) *with some $a, b > 0$ and the gradient $\nabla\mathcal{L}(k)$ is globally bounded for every $k \in \mathcal{X}$, then the flow is* GLECS.

Intuitively, the solution of a GLECS gradient flow has linear convergence away from the optimal cost, and then performs a "soft-switch" to exponential convergence once it gets near the optimal value. Fig. 1 illustrates this convergence profile and compares two different solutions, one whose cost satisfies a gPŁI and one that satisfies a *sat*PŁI.

It is easy to verify, as stated in [22], that a gPŁI implies a *sat*PŁI, but the converse is not true. In this paper, we draw attention to the fact that, through an appropriate reparameterization, a *sat*PŁI problem can be reformulated into one satisfying a gPŁI. To illustrate this, we next present a simple case where a seemingly trivial reparameterization converts a problem whose gradient flow originally only satisfied a *sat*PŁI into one that can be shown to satisfy a gPŁI.

B. Motivating example

Let $\mathcal{X} = (h, \infty)$, $h > 0$. Assume $\mathcal{L} : \mathcal{X} \to \mathbb{R}$ is real-analytic, bounded below, proper, satisfies a *sat*PŁI with constants $a, b > 0$, and is such that $\limsup_{k\to\infty} \mathcal{L}'(k)$ is finite. For this function, let $k^\star$ be the largest element of $\mathcal{T} := \{k \in (h, \infty) \mid \mathcal{L}(k) = \underline{\mathcal{L}}\}$, and define $L_\infty := \max_{k\in[k^\star,\infty)} \mathcal{L}'(k)$. It is easy to see from the results presented previously that a solution of the gradient flow for this cost function initialized at points larger than $k^\star$ would present a linear-exponential profile similar to the one described in Definition 4.

Now, consider the following reparameterization

$$f(k) := \mathcal{L}(k^2), \qquad k \geq \sqrt{h},$$

which implies that $\underline{f} := \inf_{k\geq\sqrt{h}} f(k) = \underline{\mathcal{L}}$ and $f'(k) = 2k\,\mathcal{L}'(k^2)$. It is easy to see that both optimization problems are trivially equivalent, i.e., a point k^* minimizes f if and only if $(k^*)^2$ minimizes $\mathcal{L}$, and both cost functions attain the same minimum value. Nonetheless, the gradient flow for f admits the following result:

Theorem 1. *For any $\bar{\varepsilon} > \sqrt{h}$, there exists a $\mu_{\bar{\varepsilon}} > 0$ such that for all $k \in [\bar{\varepsilon}, \infty)$, the reparameterized loss function f satisfies the following gPŁI*

$$|f'(k)| \geq \sqrt{\mu_{\bar{\varepsilon}}(f(k) - \underline{f})}\,. \tag{7}$$

Proof Sketch. A full proof is given in [26]; here we briefly outline the argument. Assume for simplicity $\underline{f} = \underline{\mathcal{L}} = 0$. We divide the proof into two parts: $k \in (\bar{\varepsilon}, k_*)$ and $k \in (k^*, \infty)$, where k_* (resp. k^*) is the smallest (resp. largest) element of $\mathcal{T} := \{k \in (\bar{\varepsilon}, \infty) \mid f(k) = 0\}$. On each region we lower bound $\psi(k) := (f'(k))^2/f(k)$ by a positive constant; taking the minimum of the two constants yields a $\mu > 0$ valid on $[\bar{\varepsilon}, \infty)$.

For $k \in (\bar{\varepsilon}, k_*)$, properness and the satPŁI imply f is strictly decreasing and $f'(k) \neq 0$, hence ψ is positive and attains a positive infimum on $(\bar{\varepsilon}, k_*)$. For $k \in (k^*, \infty)$, using $f'(k) = 2k\mathcal{L}'(k^2)$ and the satPŁI of $\mathcal{L}$ gives $\psi(k) \geq \frac{4ak^2}{b+f(k)}$, and since $k \geq k^* > 0$ this yields a uniform positive lower bound (splitting into $f(k) \leq b$ and $f(k) \geq b$). Therefore, (7) is achieved. □

By Lemma 1, (7) yields global exponential decay for the gradient flow of f, whereas the original $\mathcal{L}$ under *sat*PŁI exhibits the linear–exponential behavior of Lemma 2 for solutions initialized at values larger than $k^\star$. Thus, the reparameterization $k \mapsto k^2$ can qualitatively change the convergence behavior of solutions of a gradient flow. To further explore this phenomenon, we consider a simple and algebraically trackable case of the policy optimization LQR for a continuous-time system, whose standard formulation is known to satisfy only a *sat*PŁI and never a gPŁI [22].

III. The Overparameterized LQR Problem

This section examines how overparameterization reshapes the convergence of gradient flow solutions. To better understand the behavior of policy optimization in the overparameterized LQR problem, we study a scalar system setup shown in Problem 1 that allows for tractable analysis while preserving the essential dynamics of the general case.

Problem 1. *Consider the scalar LTI system:*

$$\dot{x} = ax + u, \quad x(0) = 1, \tag{8}$$

with an arbitrary system parameter $a \in \mathbb{R}$. Consider an overparameterized state feedback, defined as $u = -\mathbf{k}x$ where $\mathbf{k} := k_2 k_1 \in \mathbb{R}$ and $k_1, k_2^\top \in \mathbb{R}^\kappa$, for some integer $\kappa > 0$. The admissible set of control parameters (k_1, k_2) is defined as $\mathcal{K} := \{(k_1, k_2) \in \mathbb{R}^{\kappa\times 1} \times \mathbb{R}^{1\times\kappa} \mid a - \mathbf{k} < 0\}$ to ensure closed-loop stability. We define the overparameterized LQR cost $\mathcal{L} : \mathcal{K} \to \mathbb{R}$ as $\mathcal{L}(k_1, k_2) := J(\mathbf{k})$ with

$$\begin{aligned} J(\mathbf{k}) &= \mathbb{E}_{x_0\sim\mathcal{X}}\left[\int_0^\infty \left(x^2 q + u^2 r\right) dt\right] \\ &= -\frac{q + r\mathbf{k}^2}{2(a - \mathbf{k})} > 0, \end{aligned} \tag{9}$$

where $\mathcal{X} \sim \mathcal{N}(0, 1)$. Notice that stability requires $\mathbf{k} > a$, and the cost weights satisfy $q, r > 0$.

For this simple version of this problem, we can compute the gradient of $J(\mathbf{k})$ with respect to $\mathbf{k}$ explicitly as

$$\nabla J(\mathbf{k}) := \frac{r\mathbf{k}^2 - 2ar\mathbf{k} - q}{2(a - \mathbf{k})^2}, \tag{10}$$

and using the chain rule we obtain

$$\nabla_{k_1}\mathcal{L}(k_1, k_2) = \nabla J(\mathbf{k})k_2^\top, \quad \nabla_{k_2}\mathcal{L}(k_1, k_2) = \nabla J(\mathbf{k})k_1^\top.$$

Notice that the evolution of the cost along trajectories of (k_1, k_2) will satisfy

$$\dot{\mathcal{L}} := -\|\nabla\mathcal{L}(k_1, k_2)\|^2 = -\nabla J(\mathbf{k})^2(\|k_1\|^2 + \|k_2\|^2), \tag{11}$$

which is not enough to guarantee global convergence to the optimal solution. To guarantee optimality of a solution, we aim to find the largest subset of $\mathcal{K}$ in which we can establish a global Polyak–Łojasiewicz inequality (gPŁI), i.e.

$$\|\nabla\mathcal{L}(k_1, k_2)\| \geq \sqrt{\mu(\mathcal{L}(k_1, k_2) - \underline{\mathcal{L}})}, \tag{12}$$

for all $(k_1, k_2) \in \overline{\mathcal{K}}$, where $\underline{\mathcal{L}} := \inf_{(k_1,k_2)\in\mathcal{K}} \mathcal{L}(k_1, k_2)$ and $\overline{\mathcal{K}} \subseteq \mathcal{K}$.

To achieve this goal, we first define a known property of overparameterized formulations [1], [11], [25],

Definition 5 (Imbalance Measure)**.** *For the overparameterized LQR in Problem 1, the invariant matrix $\mathcal{C}$ is defined as $\mathcal{C} := k_1 k_1^\top - k_2^\top k_2 \in \mathbb{R}^{\kappa\times\kappa}$, which remains constant along any solution of the gradient flow. Furthermore, the* imbalance measure *is defined as $c := 2\,\mathrm{tr}[\mathcal{C}^2] - (\mathrm{tr}[\mathcal{C}])^2$.*

The invariant $\mathcal{C}$ is called this because it is invariant along any solution of the overparameterized gradient flow. This is a well-known property of overparameterization and is proven for the LQR in [11]. The imbalance measure c quantifies the asymmetry: it vanishes when the two factors (k_1, k_2) are perfectly aligned (up to sign) and increases as asymmetry grows. Notice that for the scalar case, the invariant $\mathcal{C}$ is the difference of two rank-one positive semidefinite matrices, for which the following result can be stated.

Proposition 1. *Let $A, B \in \mathbb{R}^{n\times n}$ be any two rank-one positive semidefinite matrices, and let $C = A - B$. Then $c := 2\,\mathrm{tr}[C^2] - (\mathrm{tr}[C])^2 \geq 0$, with $c = 0$ iff $A = B$.*

Proof. Since A, B are rank-one PSD, write $A = aa^\top$ and $B = bb^\top$ for some $a, b \in \mathbb{R}^n$. Let $\mathbf{a} := a^\top a$ and $\mathbf{b} := b^\top b$. Then notice, after some algebraic manipulation, that

$$\begin{aligned} c := 2\,\mathrm{tr}[C^2] - (\mathrm{tr}[C])^2 &= (\mathbf{a}+\mathbf{b})^2 - 4(a^\top b)^2 \\ &\geq (\mathbf{a}+\mathbf{b})^2 - 4\mathbf{a}\mathbf{b} = (\mathbf{a}-\mathbf{b})^2 \geq 0, \end{aligned}$$

concluding the proof. □

Proposition 1 guarantees that the imbalance measure c is always nonnegative. Furthermore, $c = 0$ if and only if $k_1 = \pm k_2$, which we can leverage to show that the center-stable manifold of the saddle point lies entirely within the set $\{(k_1, k_2) \in \mathcal{K} \mid c(k_1, k_2) = 0\}$ [11]. With this in mind, and with the constraint set $\mathcal{K}_\gamma$ defined below, we can state the following result.

Theorem 2. *Consider the overparameterized model-free LQR in Problem 1 and for arbitrary $\gamma > 0$, define the set $\mathcal{K}_\gamma := \big\{(k_1, k_2) \in \mathcal{K} \,\big|\, d(k_1, k_2) := \|k_1 + k_2^\top\|^2 \geq \gamma\big\}$. Then, for $\gamma > \max(0, 4a)$:*

1) *any solution initialized in $\mathcal{K}_\gamma$ remains in $\mathcal{K}_\gamma$; and*
2) *the gPŁI $\|\nabla\mathcal{L}(k_1, k_2)\| \geq \sqrt{\mu_\gamma(\mathcal{L}(k_1, k_2) - \underline{\mathcal{L}})}$ holds for all $(k_1, k_2) \in \mathcal{K}_\gamma$ with*

$$\mu_\gamma(c,\gamma) = \begin{cases} \frac{r}{4}\sqrt{\frac{4c+\gamma^2}{(\gamma-4a)^2}} > 0 & \text{if } a \geq 0, \\ \frac{r}{4}\sqrt{\frac{(\gamma^2+c)^2}{(\gamma^2-c-4a\gamma)^2}} > 0 & \text{if } a < 0, c < \tilde{c} \\ \frac{r}{4}\sqrt{\frac{c}{4a^2+c}} > 0 & \text{if } a < 0, c \geq \tilde{c} \end{cases}$$

where $\tilde{c} := \frac{a\gamma^2}{a-\gamma} > 0$.

Proof Sketch. The full derivation is provided in the arXiv version [26]; here we outline the main idea. Let $\mathbf{k}^*$ denote the optimal gain minimizing $\tilde{J}(\mathbf{k})$ and define the perturbation $\varepsilon := \mathbf{k} - \mathbf{k}^*$. After some algebraic manipulation, one obtains the gradient norm as

$$\|\nabla\mathcal{L}(k_1, k_2)\|^2 = \vartheta_1(\varepsilon)\vartheta_2(\mathbf{k})(\mathcal{L}(k_1, k_2) - \underline{\mathcal{L}})$$

where $\vartheta_1(\varepsilon)$ depends on the scalar perturbation ε and $\vartheta_2(\mathbf{k})$ depends on the imbalance c and the gain $\mathbf{k} = k_2 k_1$. For the first factor $\vartheta_1(\varepsilon)$, one shows that $\vartheta_1(\varepsilon) \geq r/4$ for all $\mathbf{k} > a$. The main task is therefore to obtain a positive lower bound for $\vartheta_2(\mathbf{k})$ on the constraint set $\mathcal{K}_\gamma$.

This is done by analyzing separately the cases $a \geq 0$ and $a < 0$. When $a \geq 0$, the saddle point introduced by overparameterization lies outside the feasible region defined by $\mathcal{K}_\gamma$, which allows us to obtain a uniform positive lower bound on $\vartheta_2(\mathbf{k})$. When $a < 0$, the geometry is more subtle and the separation induced by the threshold $\tilde{c}$ becomes important: depending on whether $c < \tilde{c}$ or $c \geq \tilde{c}$, different bounds on $\vartheta_2(\mathbf{k})$ are obtained. In both cases, the constraint $\|k_1 + k_2^\top\|^2 \geq \gamma$ ensures that $\vartheta_2(\mathbf{k})$ remains strictly positive on $\mathcal{K}_\gamma$ when $\gamma > \max(0, 4a)$. Combining the bounds on ϑ_1 and ϑ_2 yields the gPŁI with constant $\mu_\gamma(c, \gamma) > 0$. □

From Theorem 2, the constant $\mu_\gamma(c, \gamma)$ depends jointly on the imbalance measure c and the set $\mathcal{K}_\gamma$ with parameter γ, for arbitrary a. Thus, when we seek a *uniform guarantee* that the gPŁI holds for all admissible $(k_1, k_2) \in \mathcal{K}_\gamma$, we take the smallest possible value of $\mu_\gamma(c, \gamma)$ across all $c \geq 0$ for a fixed γ, yielding the following corollary.

Corollary 1. *Consider the overparameterized model-free LQR in Problem 1. For arbitrary $\gamma > 0$, define the set $\mathcal{K}_\gamma := \big\{(k_1, k_2) \in \mathcal{K} \,\big|\, d(k_1, k_2) := \|k_1 + k_2^\top\|^2 \geq \gamma\big\}$ and for all $\gamma > \max(0, 4a)$, define*

$$\underline{\mu} := \frac{r}{4}\min\left\{1, \sqrt{\frac{\gamma^2}{(\gamma-4a)^2}}\right\} > 0. \tag{13}$$

Then, it holds that $\|\nabla\mathcal{L}(k_1, k_2)\| \geq \sqrt{\underline{\mu}(\mathcal{L}(k_1, k_2) - \underline{\mathcal{L}})}$ for all $(k_1, k_2) \in \mathcal{K}_\gamma$.

Proof Sketch. See the arXiv version [26] for a full derivation. From Theorem 2, the gPŁI holds on $\mathcal{K}_\gamma$ with constant $\mu_\gamma(c, \gamma) > 0$, which depends on the imbalance $c \geq 0$. To obtain a uniform constant valid for all $(k_1, k_2) \in \mathcal{K}_\gamma$, we take $\underline{\mu} := \inf_{c\geq 0} \mu_\gamma(c, \gamma)$. Since the first factor ϑ_1 is uniformly bounded below by $r/4$ (independent of c), we evaluate the second factor ϑ_2 that depends on c. Evaluating its minimum over $c \geq 0$ yields the expression (13). In particular, when $a \geq 0$ the minimum occurs at $c = 0$, while for $a < 0$ the minimum is determined by the piecewise bounds derived in Theorem 2. Hence $\underline{\mu} > 0$ for all $\gamma > \max(0, 4a)$. □

Furthermore, the constant $\mu_\gamma(c, \gamma)$ in Theorem 2 prompts the question: *how does the imbalance of the factorization (k_1, k_2) affect the convergence rate within the admissible set $\mathcal{K}_\gamma$?* To answer this question, fix γ and compare two trajectories with equal initial cost, but different factorizations (hence different c). Intuitively, if $\mu_\gamma(c, \gamma)$ is monotonic in c (for a fixed γ), then a trajectory with greater imbalance c will change $\mu_\gamma(c, \gamma)$ and therefore affect the decay rate. The next corollary formalizes this intuition by analyzing the constant $\mu_\gamma(c, \gamma)$.

Corollary 2. *Consider the overparameterized model-free LQR in Problem 1. Let $(\phi_{k_1}(t), \phi_{k_2}(t))$ be the solution of the gradient flow (11) initialized at (k_1, k_2), and let $\mathcal{L}(\phi_{k_1}(t), \phi_{k_2}(t))$ be the corresponding cost trajectory. Suppose two initializations $(\bar{k}_1, \bar{k}_2), (\hat{k}_1, \hat{k}_2) \in \mathcal{K}_\gamma$ have imbalance measures $\bar{c}$ and $\hat{c}$, respectively, satisfying $\mathcal{L}(\bar{k}_1, \bar{k}_2) = \mathcal{L}(\hat{k}_1, \hat{k}_2)$ and $\bar{c} > \hat{c}$. Then the associated constants satisfy $\mu_\gamma(\bar{c}, \gamma) > \mu_\gamma(\hat{c}, \gamma)$, and the trajectories obey, for all $t > 0$,*

$$e^{-\mu_\gamma(\bar{c},\gamma)t}\big(\mathcal{L}(\bar{k}_1, \bar{k}_2) - \underline{\mathcal{L}}\big) < e^{-\mu_\gamma(\hat{c},\gamma)t}\big(\mathcal{L}(\hat{k}_1, \hat{k}_2) - \underline{\mathcal{L}}\big),$$

showing that greater imbalance accelerates convergence.

Proof Sketch. A full derivation appears in [26]. There it is shown that the constant $\mu_\gamma(c, \gamma)$ obtained in Theorem 2 is monotonic in c for any fixed γ. Therefore, for two initializations $(\bar{k}_1, \bar{k}_2)$ and $(\hat{k}_1, \hat{k}_2)$ in $\mathcal{K}_\gamma$ with equal cost but imbalance $\bar{c} > \hat{c}$, we have $\mu_\gamma(\bar{c}, \gamma) > \mu_\gamma(\hat{c}, \gamma)$, yielding the stated inequality in Corollary 2. □

The next section presents numerical experiments indicating that the formulation-dependent qualitative change from GLECS to GECS observed in the scalar model also persists for general LQR instances.

IV. Numerical Results

In this section we present numerical simulations illustrating the theoretical results in a more general setting. Our aims are twofold: (i) to demonstrate the *qualitative* change proven for the scalar case in Theorem 2; and (ii) to illustrate the trade-off between the formulations generated by the interplay of this result and the saddle points introduced by the overparameterized formulation.

We consider two continuous-time LTI systems: a stable system $\mathcal{G}_1$ and an unstable system $\mathcal{G}_2$, given by

$$\mathcal{G}_1 := \{\dot{x}_1 = A^- x_1 + B_1 u_1\}, \quad \mathcal{G}_2 := \{\dot{x}_2 = A^+ x_2 + B_2 u_2\},$$

where $x_1, x_2 \in \mathbb{R}^n$ are the states with initial conditions $x_{1,2}(0) \sim \mathcal{N}(0, I_n)$ and I_n is the identity matrix of size n. Both $u_1 \in \mathbb{R}^{m_1}$ and $u_2 \in \mathbb{R}^{m_2}$ are the control inputs given by the overparameterized neural network. Both inputs are written as $u = \mathbf{K}x$ (LFFNN), with $\mathbf{K} = K_2 K_1$, $K_1 \in \mathbb{R}^{\kappa \times n}$, $K_2 \in \mathbb{R}^{m \times \kappa}$ where m matches the number of inputs, either m_1 or m_2, and $\kappa = 10$ defines the number of neurons. The matrices are defined as $A^- := -(5I_n + A) \in \mathbb{R}^{n \times n}$ and $A^+ := A \in \mathbb{R}^{n \times n}$, with

$$A = \begin{bmatrix} 0.2373 & 0.3452 & 0.6653 & 0.6715 & 0.3288 \\ 0.3452 & 0.4889 & 0.8060 & 0.3889 & 0.5584 \\ 0.6653 & 0.8060 & 0.0377 & 0.5735 & 0.5100 \\ 0.6715 & 0.3889 & 0.5735 & 0.3354 & 0.6667 \\ 0.3288 & 0.5584 & 0.5100 & 0.6667 & 0.4942 \end{bmatrix},$$

while $B_1 \in \mathbb{R}^{n \times m_1}$ and $B_2 \in \mathbb{R}^{n \times m_2}$, denoted as

$$B_1^\top = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad B_2 = I_n,$$

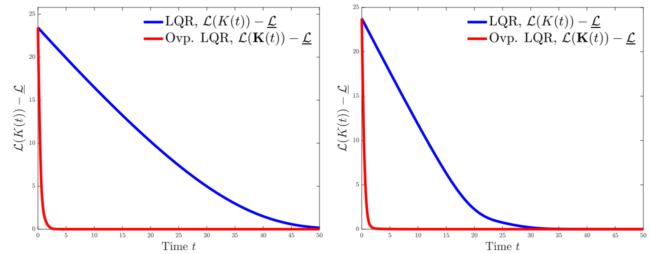
and $m_2 = n$.

We consider the overparameterized LQR problem as defined in [11], with $J(\mathbf{K})$ being the LQR cost and $\mathcal{L}(K_1, K_2) := J(\mathbf{K})$ being its overparameterized formulation. The evolution of $\mathcal{L}$ along (K_1, K_2) is defined by the following dynamics

$$\dot{K}_1 = -K_2^\top \nabla J(\mathbf{K}), \quad \dot{K}_2 = -\nabla J(\mathbf{K}) K_1^\top.$$

For both systems, $\mathcal{G}_1$ and $\mathcal{G}_2$, we take initial feedback matrices $\mathbf{K}^-(0) \in \mathbb{R}^{m_1 \times n}$ and $\mathbf{K}^+(0) \in \mathbb{R}^{m_2 \times n}$, each guaranteeing that their respective closed-loop system is stable. In the Ovp. LQR setting, these initial gains are factorized as $\mathbf{K}(0) = K_2(0)K_1(0)$, with both $K_1(0)$ and $K_2(0)$ chosen to match the initialization of each scenario, $\mathcal{G}_1$ and $\mathcal{G}_2$ (see [11] or Remark 1 as an example of how to factorize $\mathbf{K}(0)$). The two initializations are

$$\mathbf{K}^-(0) = \begin{bmatrix} -2.14 & -2.62 & 20.48 & -1.55 & -1.30 \\ -2.07 & -0.80 & -1.55 & 19.14 & -1.94 \\ -0.64 & -1.50 & -1.30 & -1.94 & 18.44 \end{bmatrix}$$



(a) Stable case $\mathcal{G}_1$ **(b)** Unstable case $\mathcal{G}_2$

Fig. 2: Optimality–gap comparison for standard LQR vs. Ovp. LQR on two systems, $\mathcal{G}_1$ and $\mathcal{G}_2$. Both methods start with the same initial cost $\mathbf{K}^-(0) = K^-(0)$ and $\mathbf{K}^+(0) = K^+(0)$ with initial states $x_1(0)$ and $x_2(0)$ in turn.

$$\mathbf{K}^+(0) = \begin{bmatrix} 12.21 & 1.12 & 2.16 & 2.18 & 1.07 \\ 1.12 & 13.03 & 2.61 & 1.26 & 1.81 \\ 2.16 & 2.61 & 11.56 & 1.86 & 1.65 \\ 2.18 & 1.26 & 1.86 & 12.53 & 2.16 \\ 1.07 & 1.81 & 1.65 & 2.16 & 13.02 \end{bmatrix}.$$

The optimal value of the cost is denoted by $\underline{\mathcal{L}}$, while the optimality gap is defined as $\mathcal{L}(\mathbf{K}(t)) - \underline{\mathcal{L}}$ for the Ovp. LQR and $\mathcal{L}(K(t)) - \underline{\mathcal{L}}$ for the standard LQR.

We discuss the first point: the Ovp. LQR exhibits GECS, whereas the standard LQR shows GLECS. This is illustrated in Figs. 2a–2b, which compare the optimality gap of the standard LQR policy optimization (blue) with its Ovp. LQR counterpart (red) for the stable system $\mathcal{G}_1$ and the unstable system $\mathcal{G}_2$. Consistent with Theorem 2 and Corollary 1, the Ovp. LQR exhibits exponential decay of the cost gap (GECS), while the standard LQR shows the characteristic linear–exponential profile (GLECS). The effect appears in both the stable and unstable regimes, underscoring that the qualitative improvement arises from the reparameterization rather than system stability. Empirically, the factorization (K_1, K_2) reshapes the geometry so that the policy gradient through the chain rule can be larger in descent directions of K, yielding faster convergence.

Figure 3 shows the second point using $\mathcal{G}_1$: what happens when the Ovp. initialization is close to the center–stable manifold of the saddle ($\gamma \approx 0$, hence $c \approx 0$), where we pick $\mathbf{K}(0) \approx 0$. According to Theorem 2 and Corollary 1, this results in a small value of the constant $\mu_\gamma(c, \gamma)$, yielding slow convergence (often slower than the standard LQR) for the gradient flow but still consistent with exponential decay. Practically, this highlights a trade–off: overparameterization yields GECS, but care is needed to avoid the saddle (e.g., $d \geq \gamma$ at initialization) to prevent slow convergence.

Remark 1 (Initialization of $K_1(0)$ and $K_2(0)$)**.** *Define the initial cost gap $\eta > 0$. Let K^* be the LQR gain for (A, B, Q, R). Choose a scale $s > 0$ and set $\mathbf{K}^* := sK^*$. Accept $\mathbf{K}^*$ only if $A_{\mathrm{cl}} := A - B\mathbf{K}^*$ is Hurwitz, in which case the cost $\mathcal{L}(\mathbf{K}^*) = \mathrm{trace}(P_0)$ where P_0 solves*

$$A_{\mathrm{cl}}^\top P_0 + P_0 A_{\mathrm{cl}} + \left(Q + \mathbf{K}^{*\top} R \mathbf{K}^*\right) = 0,$$

yielding the gap $\theta_0 := \mathcal{L}(\mathbf{K}^*) - \underline{\mathcal{L}}$. Increase s geometrically until the gap exceeds the target $\theta_0 \geq \eta$, then set $\mathbf{K}(0) := \mathbf{K}^*$.

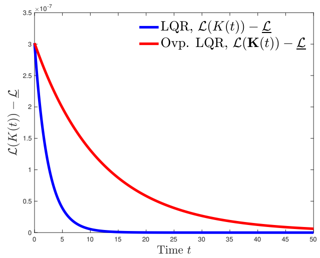


Fig. 3: The optimality–gap in $\mathcal{G}_1$ using $\mathbf{K}^-(0) = K^-(0) \approx 0_{m,n}$

Draw any full-column-rank matrix $K_1(0) \in \mathbb{R}^{\kappa \times n}$ with $\kappa \geq n$ and define $K_2(0) := \mathbf{K}(0)(K_1(0)^\top K_1(0))^{-1} K_1(0)^\top$. Hence $K_2(0)K_1(0) = \mathbf{K}(0)$. This initialization (i) ensures a controlled initial gap relative to the optimum and (ii) matches the desired overparameterized product exactly.

V. Conclusion and Future Work

We proved that problem formulation shapes the convergence behavior of gradient flows in nonconvex optimization. A simple reparameterization $k \mapsto k^2$ converts a landscape that satisfies only a *sat*PŁI into one that satisfies a *g*PŁI *almost* everywhere, showing a *qualitative* change in convergence for the solution; from global linear–exponential cost stability (GLECS) to global exponential cost stability (GECS). Motivated by this observation, we considered the policy-optimization LQR as a testbed and showed that an overparameterized (LFFNN) policy achieved the same *qualitative* upgrade. For the scalar testbed, we characterized a *g*PŁI with an *explicit* rate constant that depends on an "imbalance" measure $c \geq 0$ and an invariant region $\mathcal{K}_\gamma$ in which solutions must be initialized.

Numerical experiments in a general setting on both stable and unstable systems confirmed the theory: the overparameterized formulation exhibits GECS while the standard formulation shows GLECS, and highlighted the expected slow convergence only when initialized near the saddle manifold ($c \approx 0$). Future work will extend these guarantees to stochastic gradients and noisy dynamics; generalize beyond LQR to broader control/learning problems; and analyze deeper or alternative factorizations to sharpen *g*PŁI constants and robustness.

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